

Digestive system

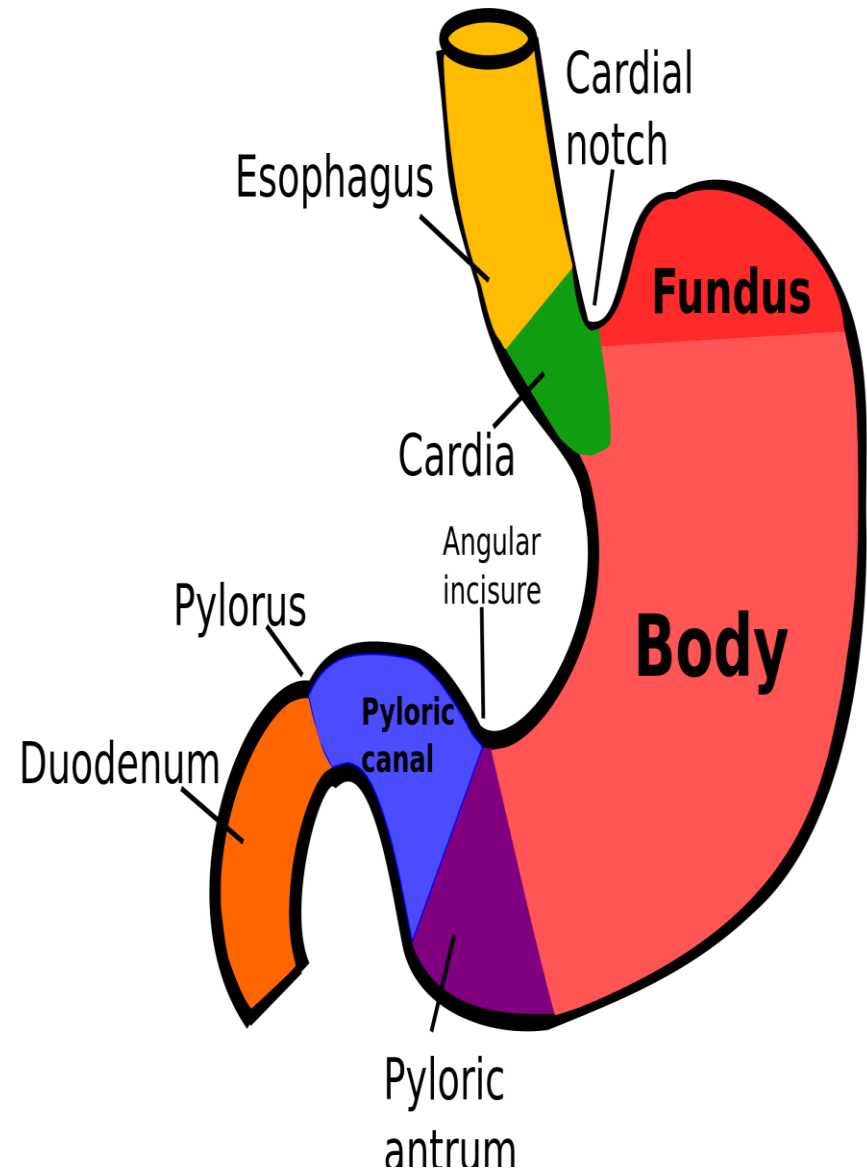
Histological structures of different regions of gastric mucosa

LG 2

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Stomach divided into 4 regions:

- **1-Cardic**: extend from the squamocolumnar junction at the lower end of the esophagus into upper stomach.
- **2- Fundus** – dome-shaped region beneath the diaphragm
- **3- Body** – mid portion of the stomach
- **4- Pyloric** region – is continuous with the duodenum through the pyloric sphincter
- **fundus and body are identical histologically**
- **mucosa & sub- mucosa of undistended lie in longitudinally folds called (rugae) when the stomach is filled with food these fold becomes flattens.**

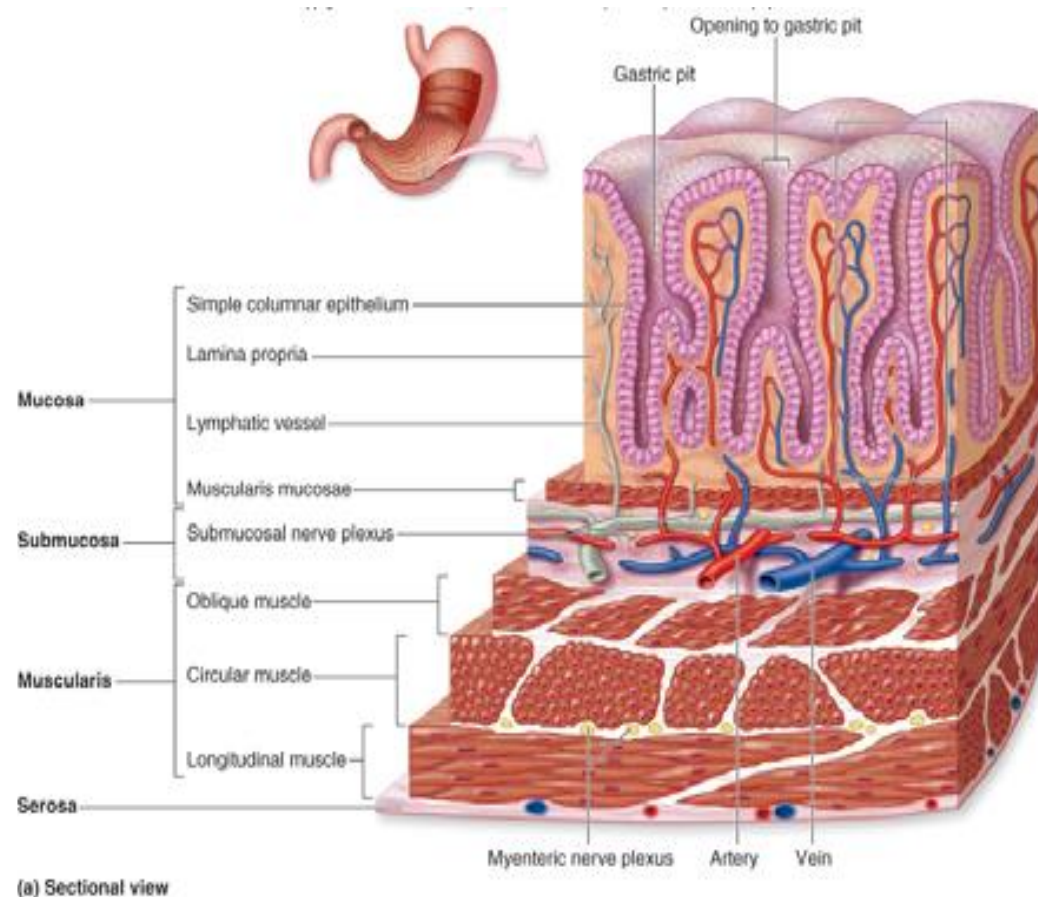


General structure of stomach:

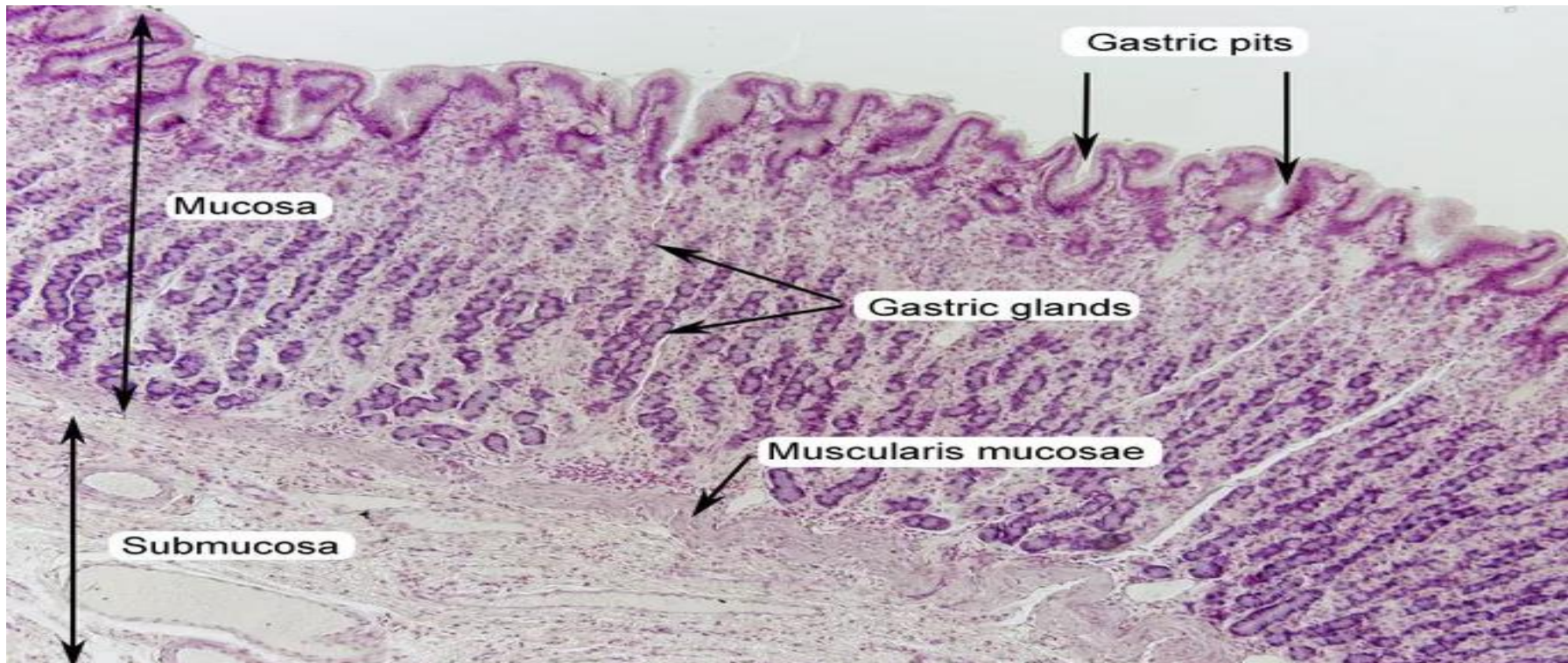
- The stomach wall has the same layers as the rest of tract:
- **mucosa**
- **sub-mucosa**
- **muscularis externa**
- **serosa**

1-The mucosa :consist of

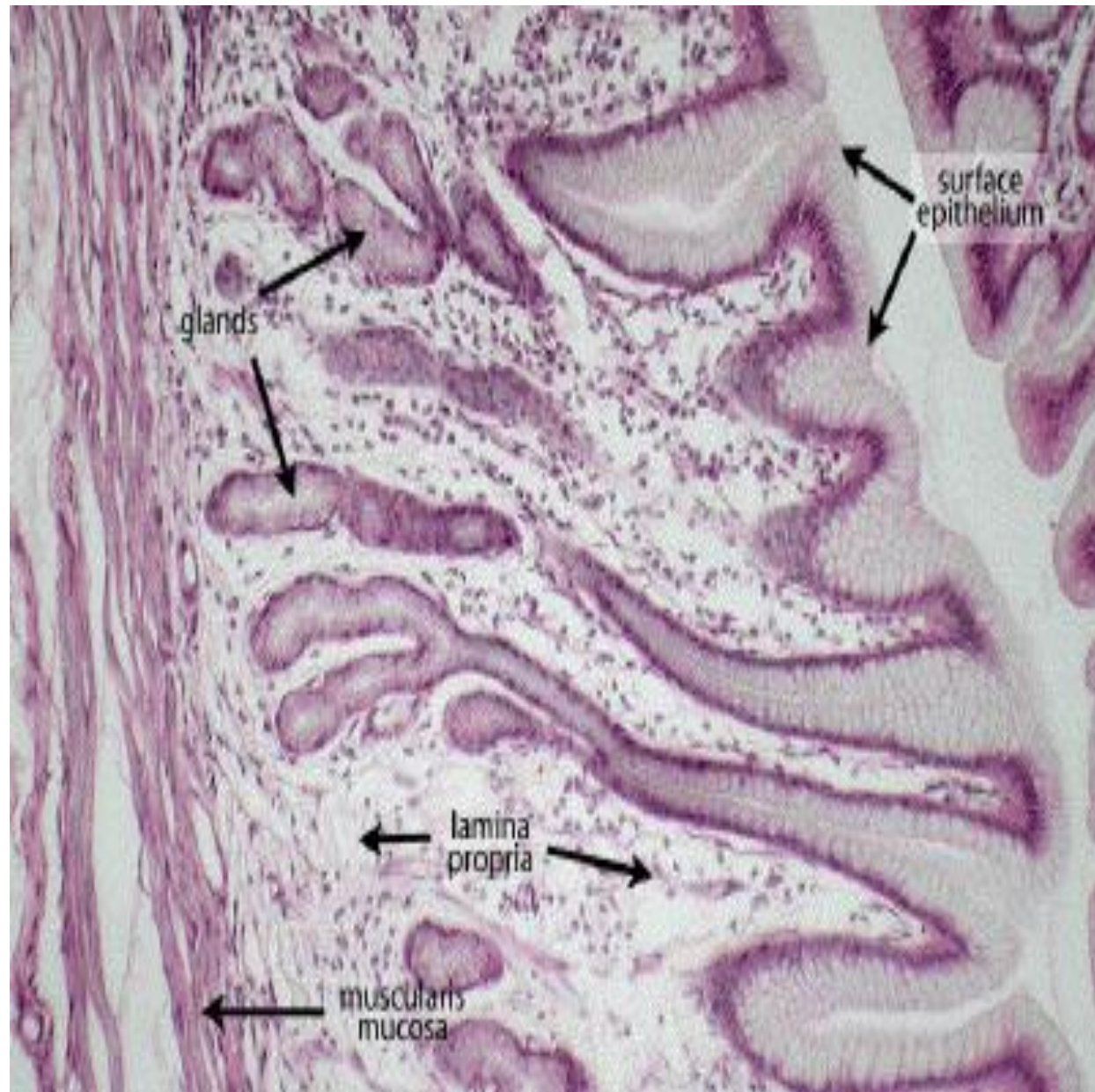
- **a-Epithelium**
- **b-lamina propria**
- **c- muscularis mucosa ,**



- **a- epithelium:** that invaginate into lamina propria forming gastric pits emptying into gastric pits are (**cardiac, gastric & pyloric**) glands.
- the epithelium covering surface & lining the pits is simple columnar ep., all cells secrete mucus form thick layer that protect ep., from the effect of strong acid secreted , by stomach tight junction around surface & pit cells also form part of barrier to acid .

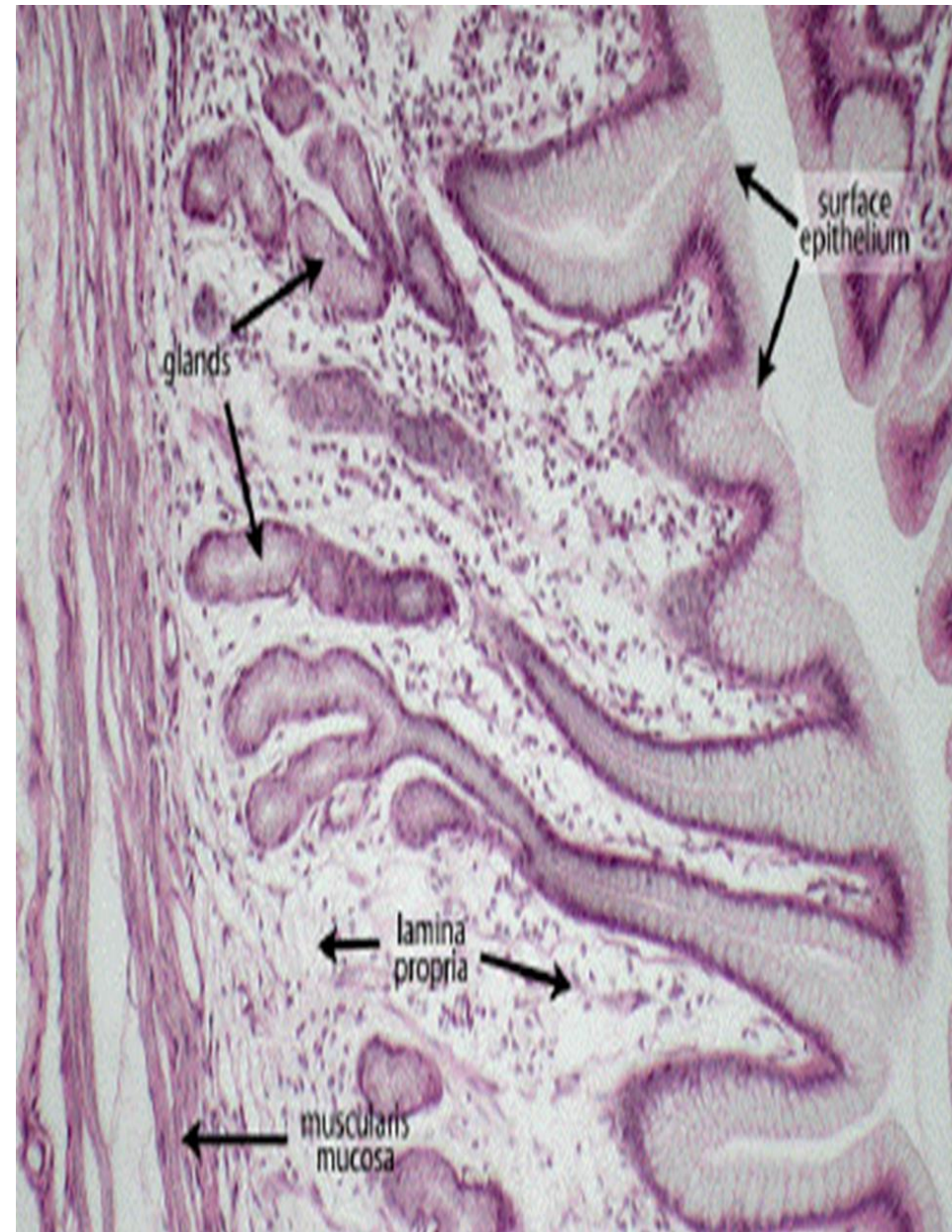


- **b-The lamina propria:** composed of loose c.t. contain smooth muscle, lymphoid cells, glands
- **c- muscularis mucosa:** smooth muscle cells that produce local movements of mucosa



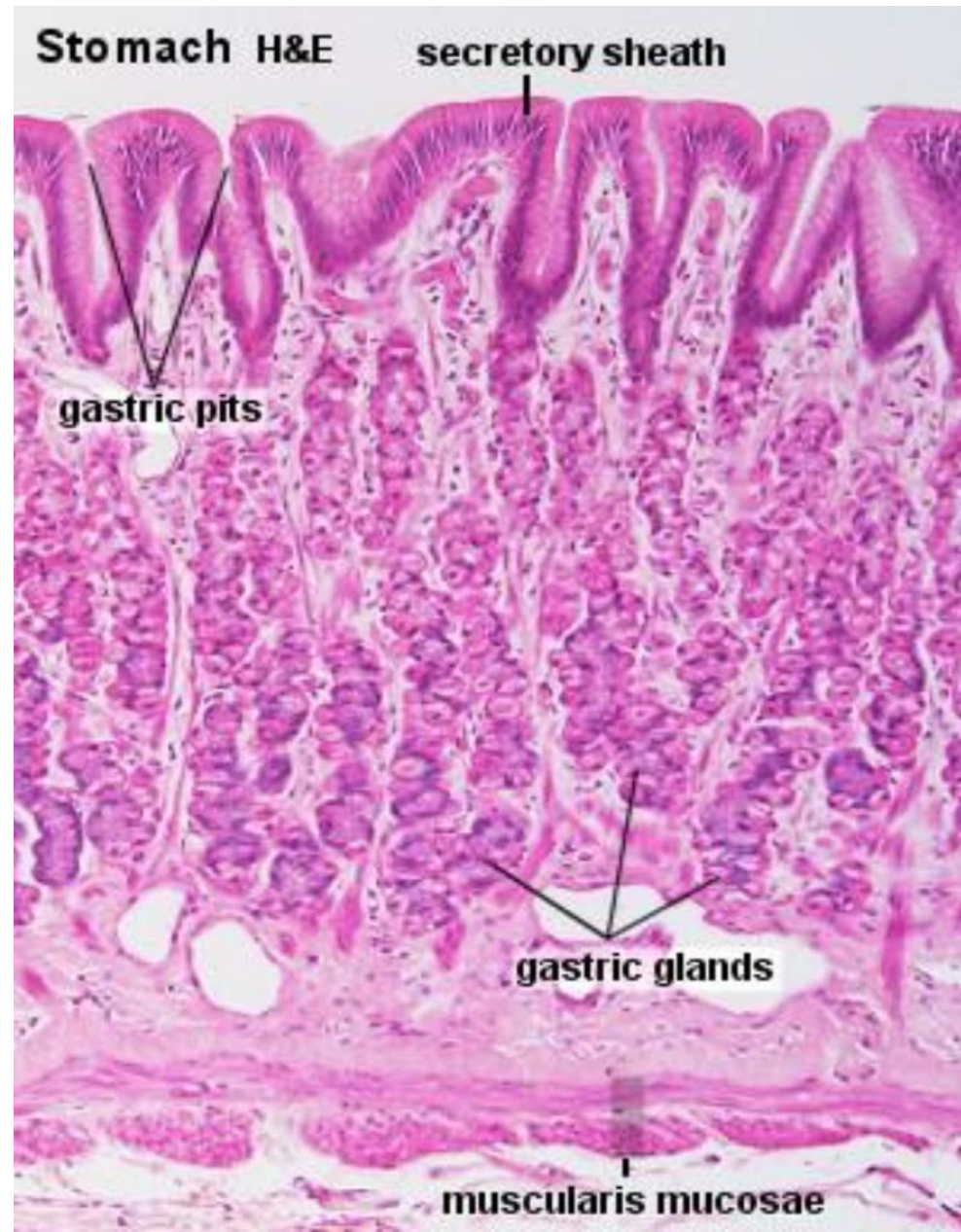
Cardiac region

- are found in a small area near the opening of the esophagus
- Lamina propria contain branched tubular **cardiac gland**
with shallow pits and glands of equal length
- most of secretory cells produce mucus and few parietal cells can be found

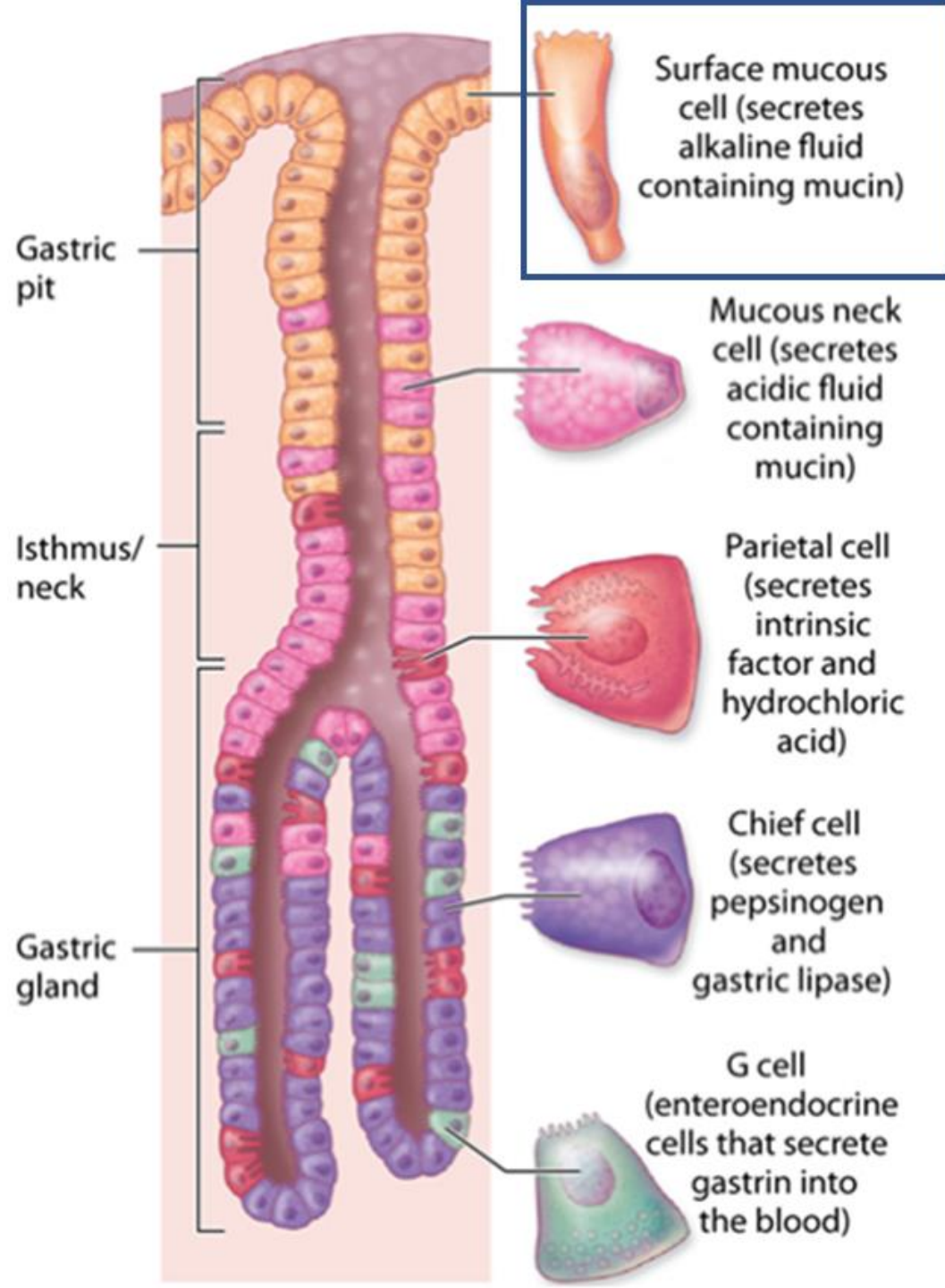


Fundus or body region:

- Is the largest region extend from cardia to pylorus
- The lamina propria of fundus & body is filled with branched tubular **gastric glands** (fundic gland) The glands open into gastric pits, each pit receive the opening of several glands
- ,these glands are shallow pits & long glands .

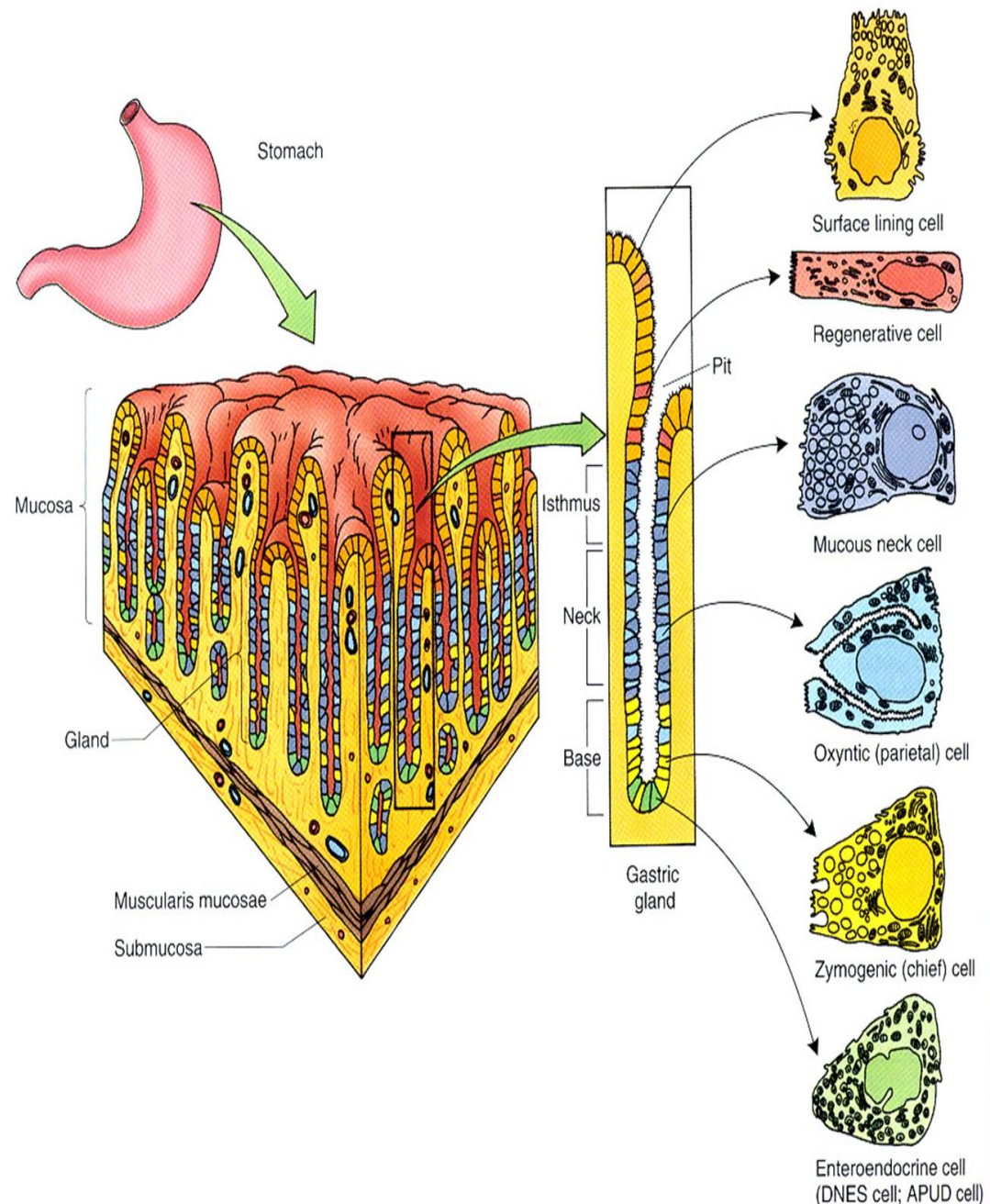


- The distribution of epithelial cells in gastric gland is not uniform:
- The neck of the gland consist of stem cells, mucous neck cells & oxyntic (parietal) cells
- The base of the glands contain few parietal cells, chief (zymogenic) cells& enteroendocrine cells



1-Stem cells

- Found in the neck region
- Are low columnar cells , oval nuclei near the base of the cells
- The cells have a high rate of mitosis, which have a turnover time 4-7 days,
- They increase in number when the gastric epithelium is damaged (for example when there is a gastric ulcer) & play an important role in healing.

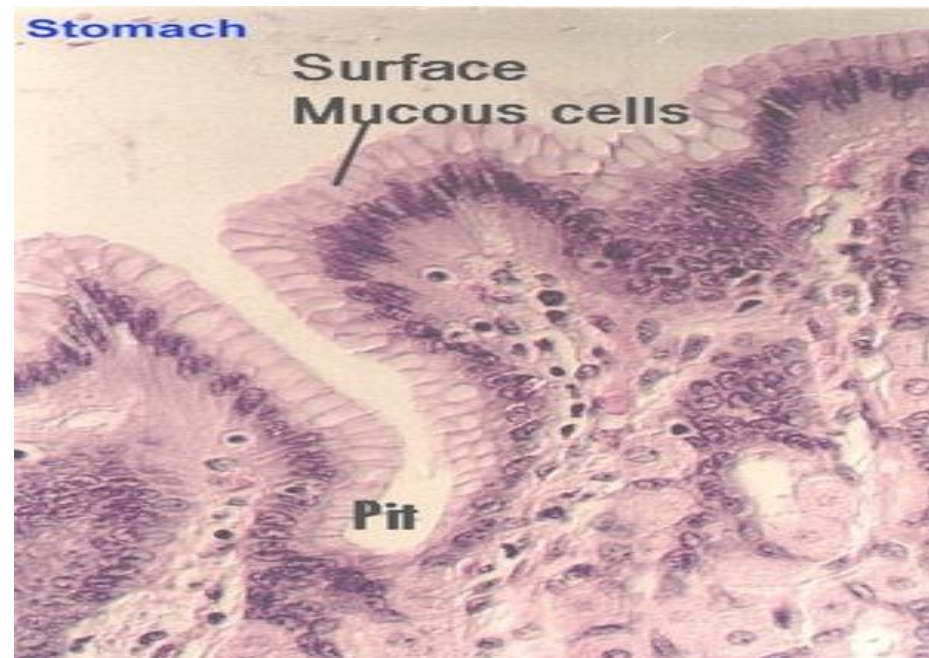
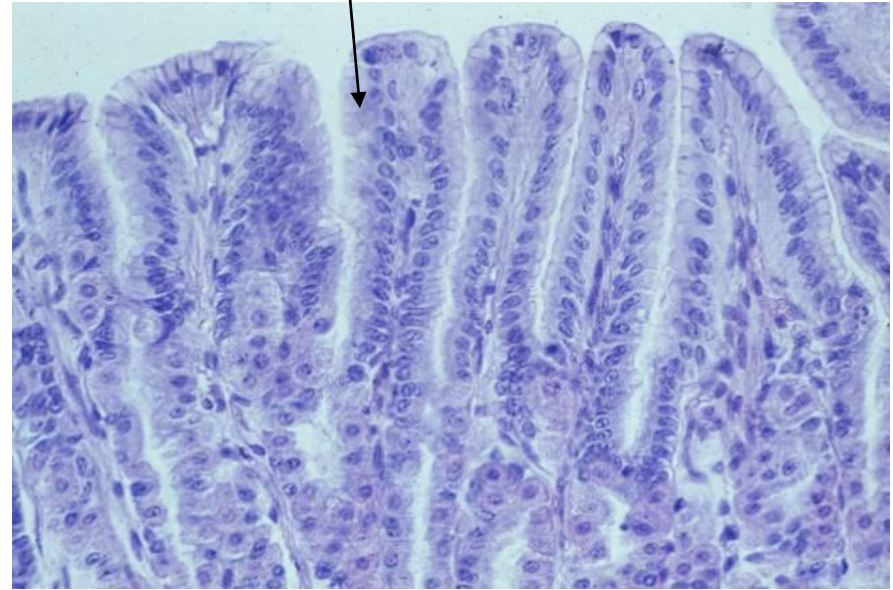


2-Mucous cells

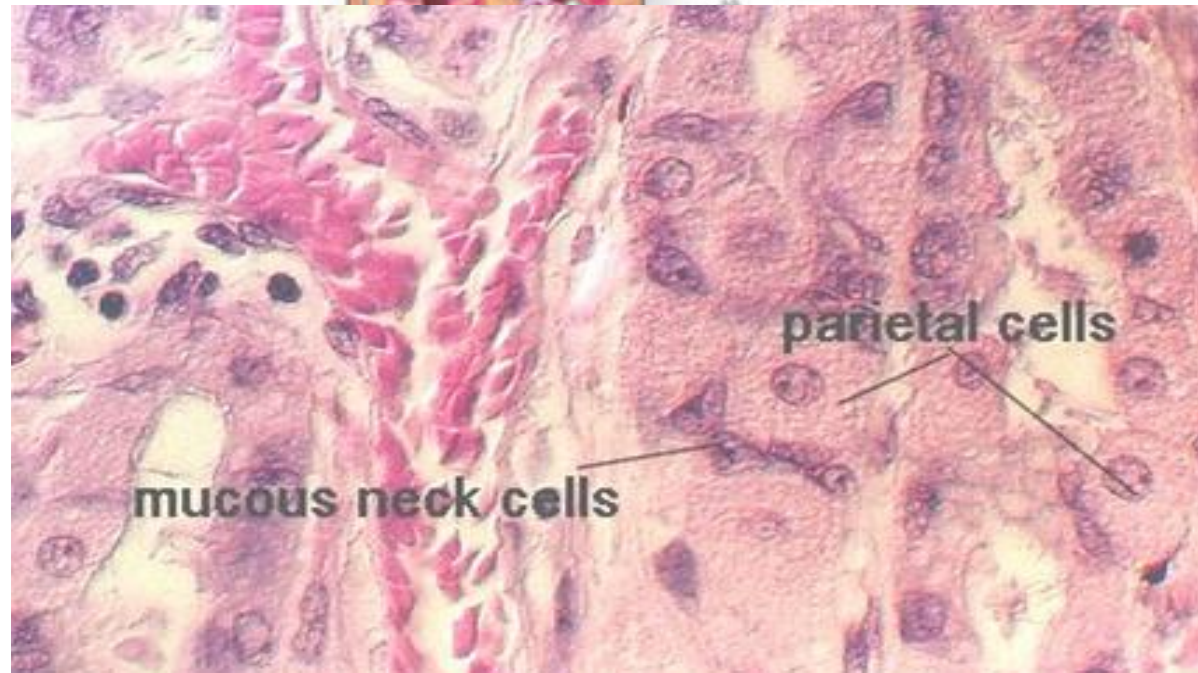
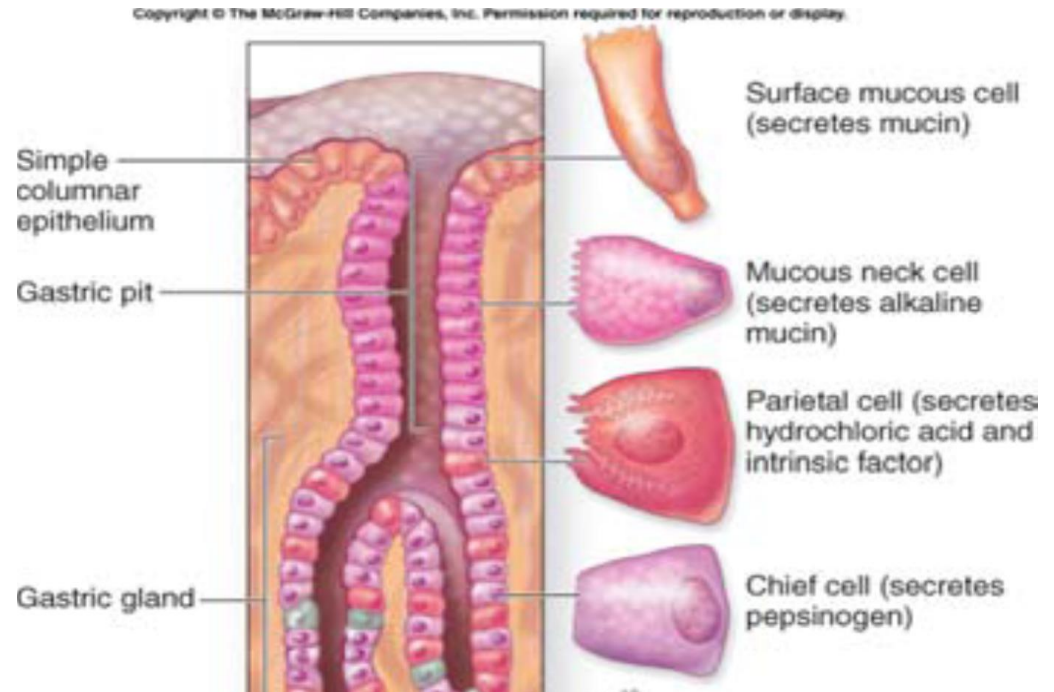
- **Are 2 types:**
- **1-surface mucous cells**
- **2- mucous neck cells**

- **1-surface mucous cells** are specialized epithelial cells lining the stomach's surface and gastric pits
- are tall columnar, basal nuclei, clear staining cytoplasm contain small mucin vacuoles, and all organelles
- Has short microvilli , the lateral cells are separated by gap junction
- They secrete neutral mucus that protect the stomach surface from acidity of gastric fluid

surface m.c

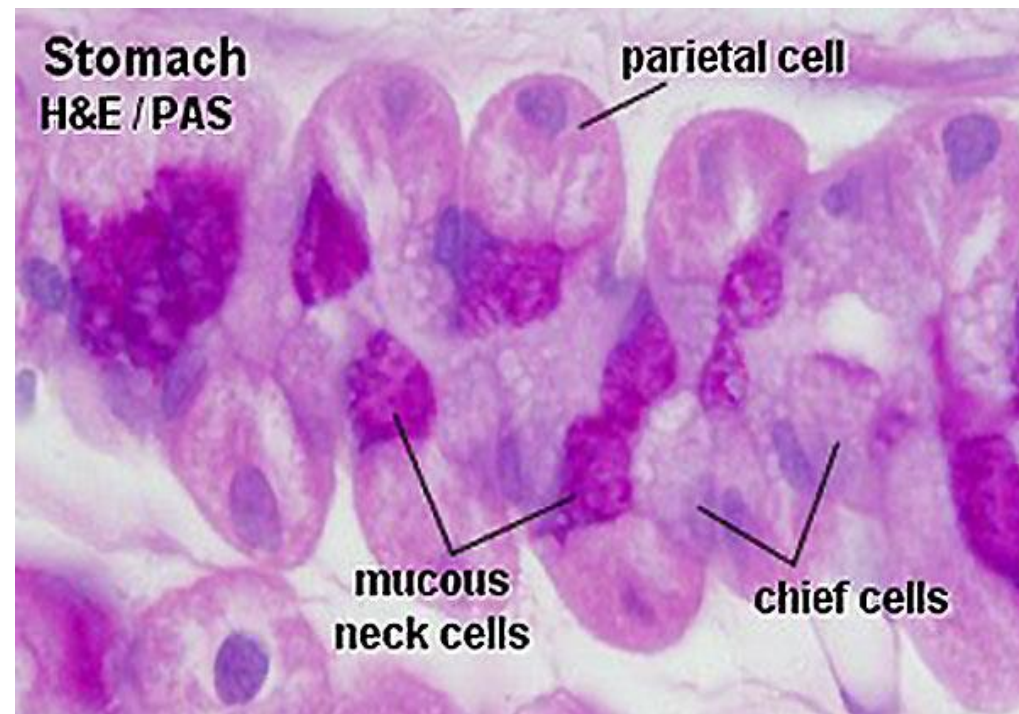
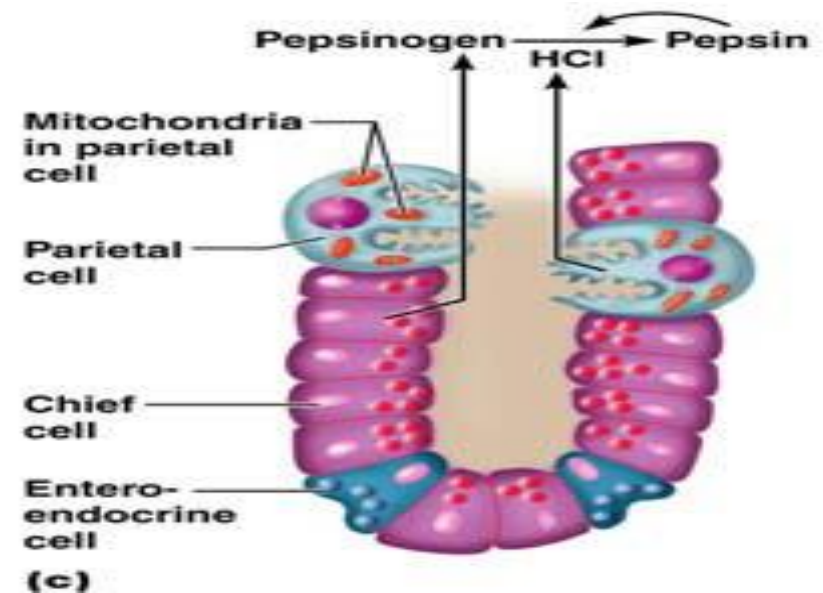


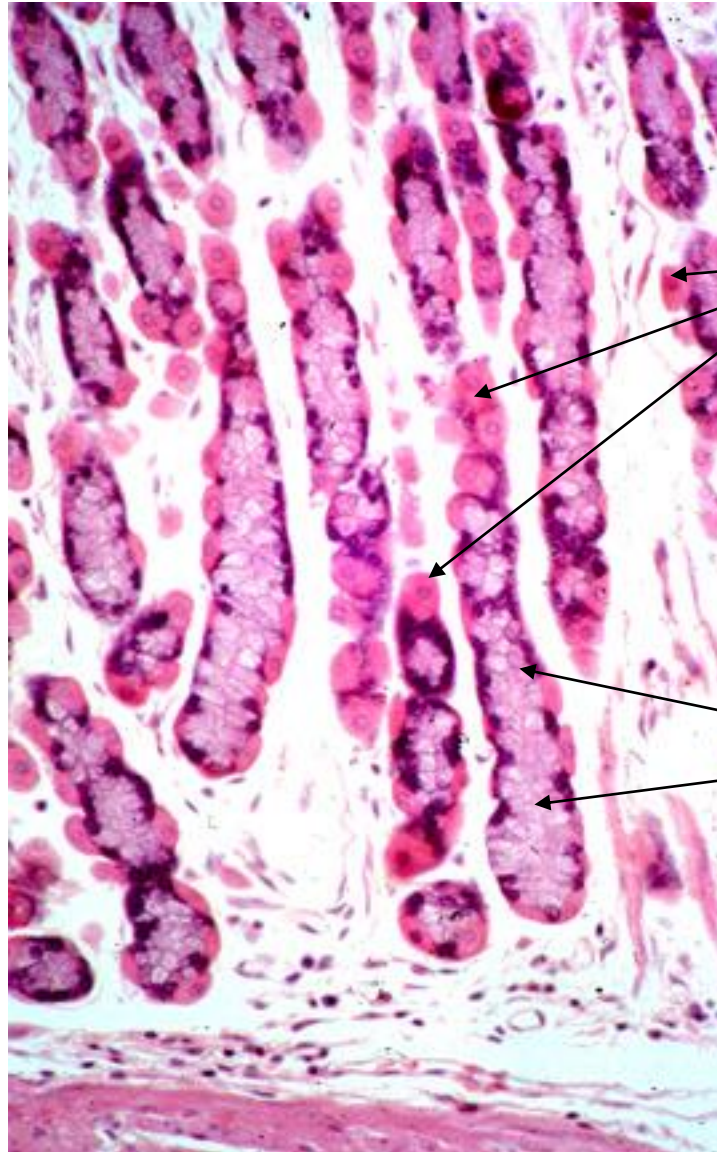
- **2- mucous neck cells** Are present in clusters or as single between parietal cells in the neck of gastric gland
- Are irregular in shape , the nucleus at the base of the cell , their secretory granules are near the apical surface
- **Their mucus secretion is less alkaline than that of the surface epithelial mucous cells.**



3-parietal(oxyntic) cells

- Are located between mucous neck cells in the neck of the gland , few in basal part
- Are large, pale & round to pyramidal in shape, have 1-2 central nuclei , acidophilic cytoplasm contain many mitochondria
- **Responsible for :**
- HCL production
- gastric intrinsic factor =is a glycoprotein required for vitamin B12 absorption

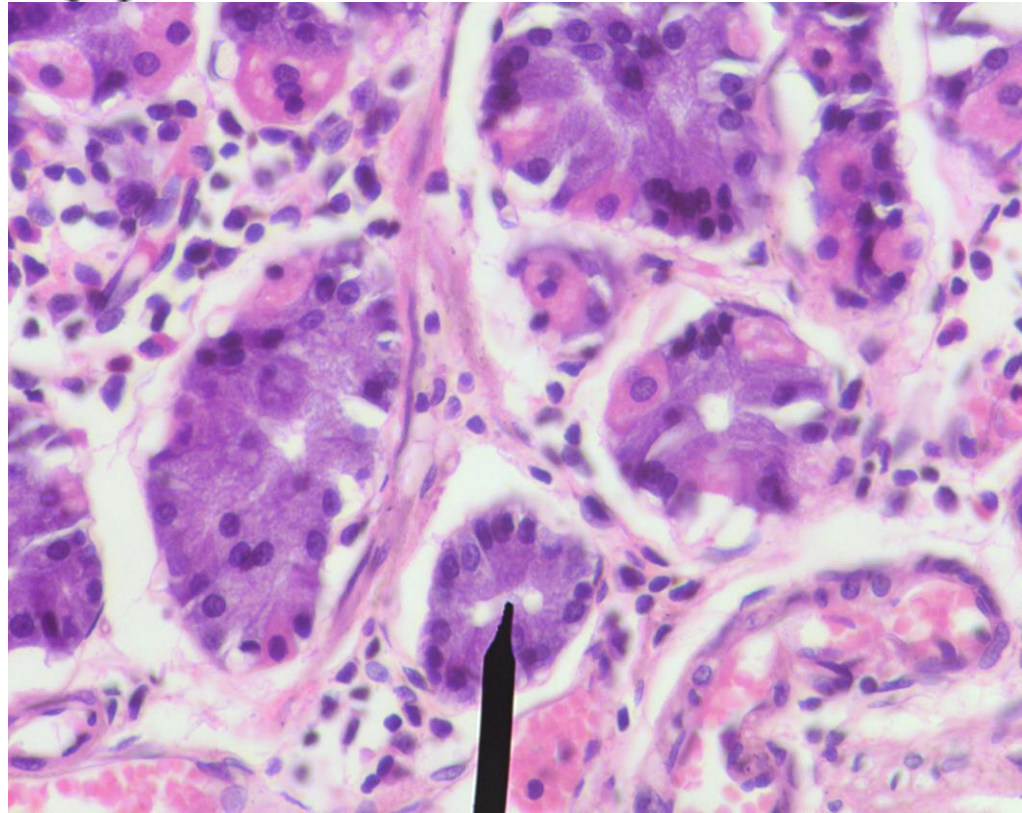
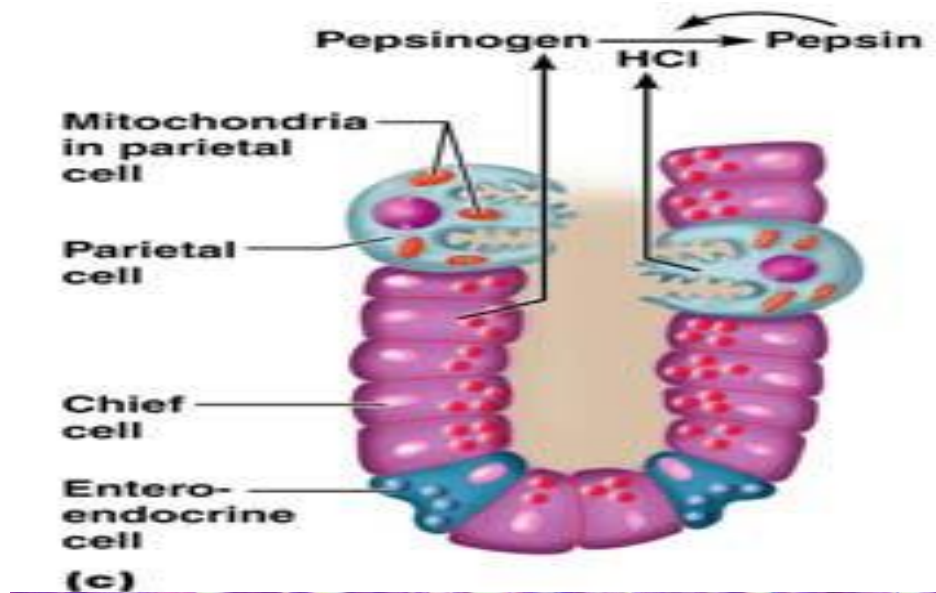




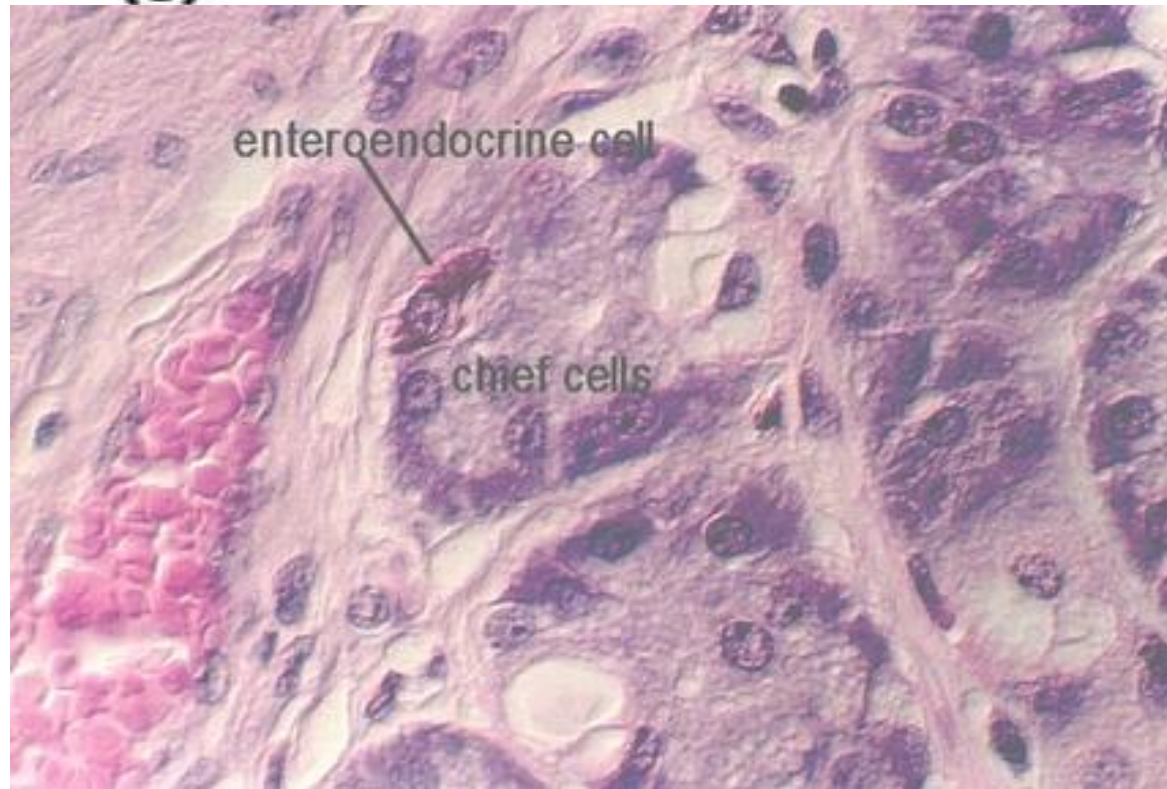
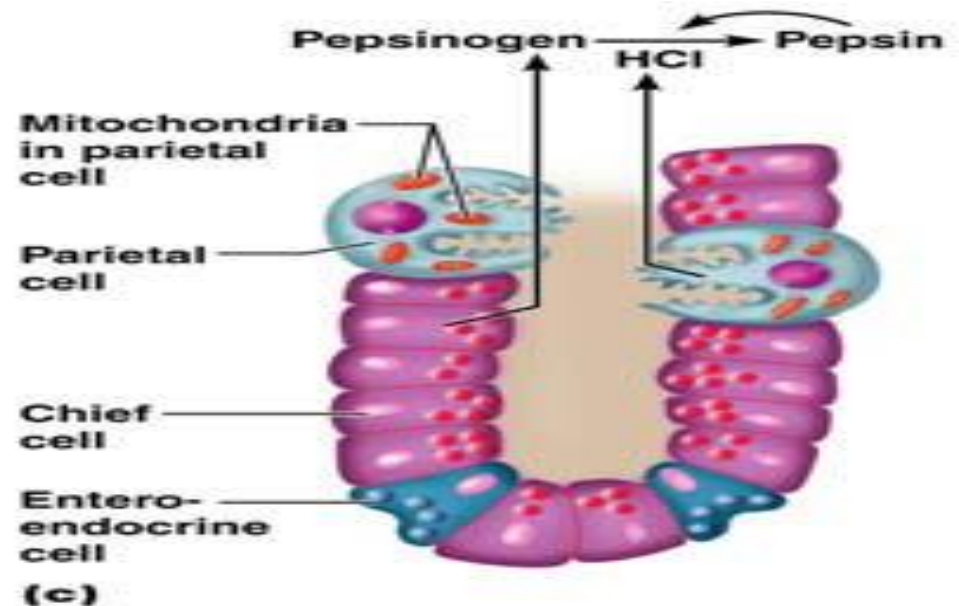
Parietal cell

Chief cell

- **4-Chief (zymogenic) cells:**
Are predominant in the base of the gastric glands
- Are smaller than parietal cells
- Are basophilic cells contain many RER , ribosomes,
- they also contain membrane limited pepsinogen granules the granules contain inactive enzyme pepsinogens precursors which are converted in the acid of the stomach into active pepsins.
- Pepsins initiate the hydrolysis of ingested protein in the stomach.
- Chief cells also produce gastric lipase, which digests many lipids.



- **Enteroendocrine cells (APUD cells):** Located in the basal part of fundic gland
 - Are small, round
 - clear cytoplasm, contain membrane bound neurosecretory granules
- ‘secrete:
- gastrin
 - endorphins
 - serotonin



Pylorus region

- This make up the distal 4-5cm. of the stomach , leading to the small intestine
- Lamina propria contain **pyloric gland** have long pit & short gland
- mucus –secreting cells are predominant , chief cells are scarce , G –cells which release gastrin are found &D-cells secrete somatostatin

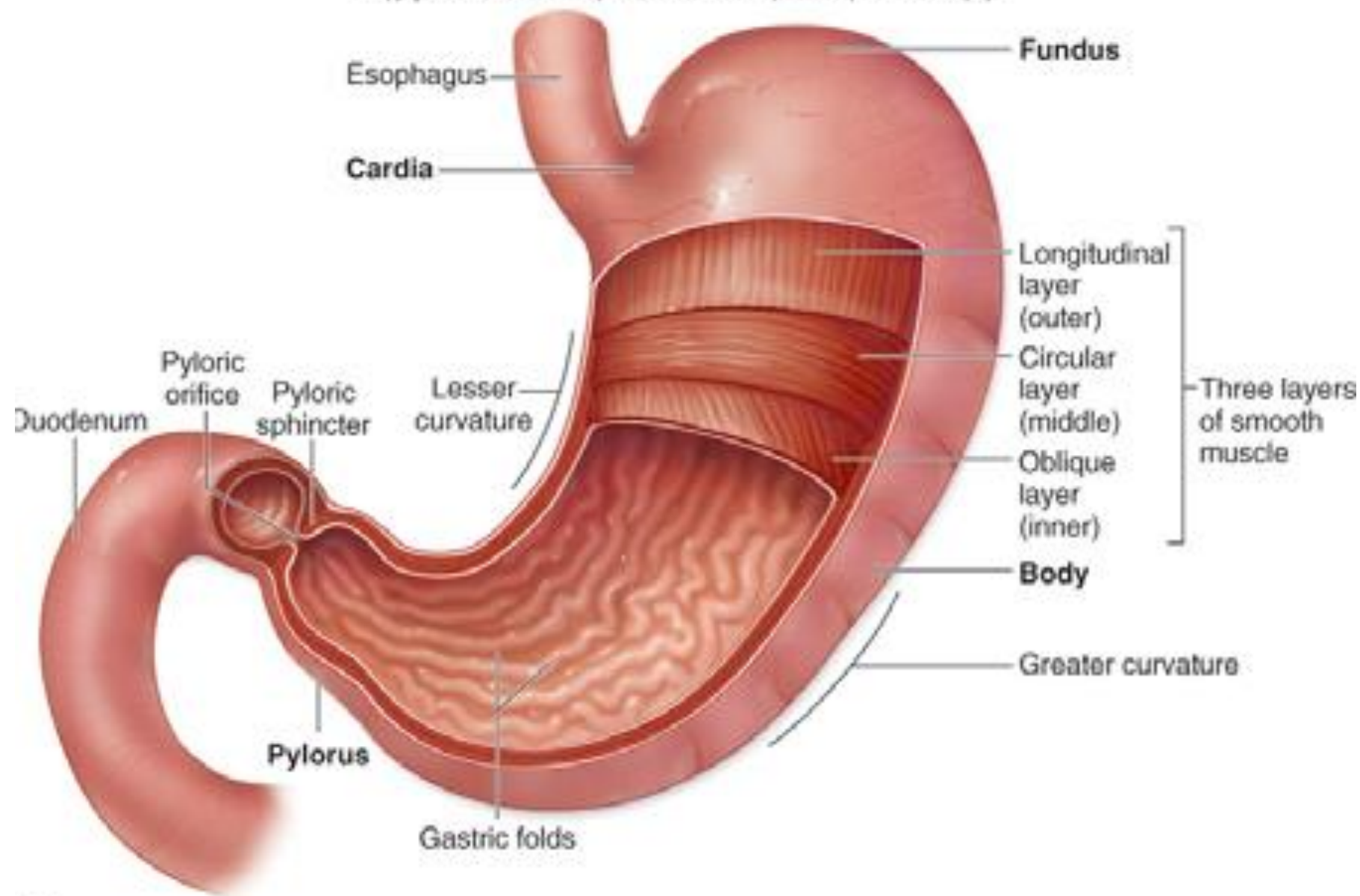
2-Sub-mucosa:

3-Mscularis :smooth muscle fibers,orianted in 3 direction :
outer longitudinal, middle circular inner oblique

At the pyloric region the middle layer is thickened to form the **pyloric sphincter**

4-Serosa:Is thin , covered by mesothelium





(a)

APUD CELLS IN GIT tract

1-D cells Pylorus, duodenum, and pancreatic islets **Somatostatin** **Secretion from other DNES**

2-EC cells Stomach, small and large intestines **Serotonin and substance P**
Increased gut motility

3-G cells Pylorus **Gastrin** Gastric acid secretion

4-I cells Small intestine **Cholecystokinin (CCK)** Pancreatic enzyme secretion, gallbladder contraction **Gastric acid secretion**

5-K cells Duodenum and jejunum **Gastric inhibitory polypeptide (GIP)** **Gastric acid secretion**

6-L cells Ileum and colon **Glucagon-like peptide** Insulin secretion **Gastric acid secretion** **Sense of hunger**

7-L cells Ileum and colon **Peptide YY** H₂O and electrolyte absorption in large intestine **Gastric acid secretion**

8-Mo cells Small intestine **Motilin** Increased gut motility

9-N cells Ileum **Neurotensin** **Gastric acid secretion**

10-S cells Small intestine **Secretin** Pancreatic and biliary bicarbonate and water secretion **Gastric acid secretion**

Thanks